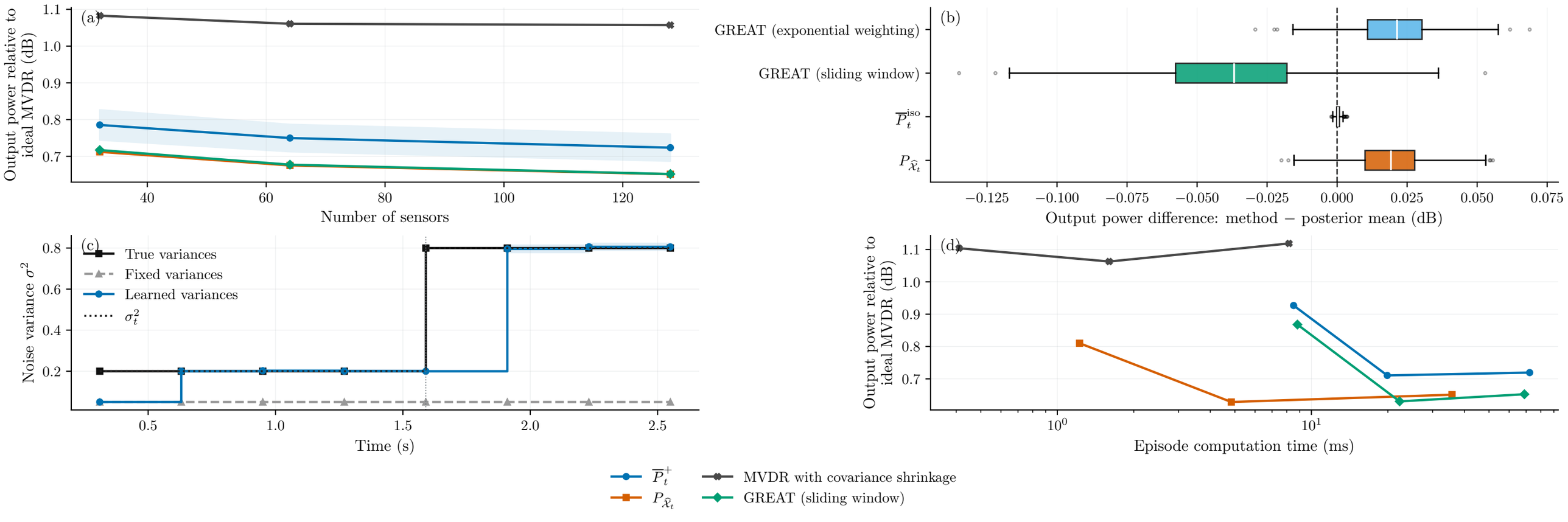

(a)
Output power relative to ideal MVDR (dB)
1.1
1.0
0.9
0.8
0.7
40
60
80
100
120
Number of sensors
(b)
GREAT (exponential weighting)
GREAT (sliding window)
$\overline{P}_t^{\text{iso}}$
$P_{\widehat{\mathcal{X}}_t}$
−0.125
−0.100
−0.075
−0.050
−0.025
0.000
0.025
0.050
0.075
Output power difference: method − posterior mean (dB)
(c)
Noise variance σ^2
0.8
0.6
0.4
0.2
True variances
Fixed variances
Learned variances
σ_t^2
0.5
1.0
1.5
2.0
2.5
Time (s)
(d)
Output power relative to ideal MVDR (dB)
1.1
1.0
0.9
0.8
0.7
10^0
10^1
Episode computation time (ms)
$\overline{P}_t^+$
$P_{\widehat{\mathcal{X}}_t}$
MVDR with covariance shrinkage
GREAT (sliding window)